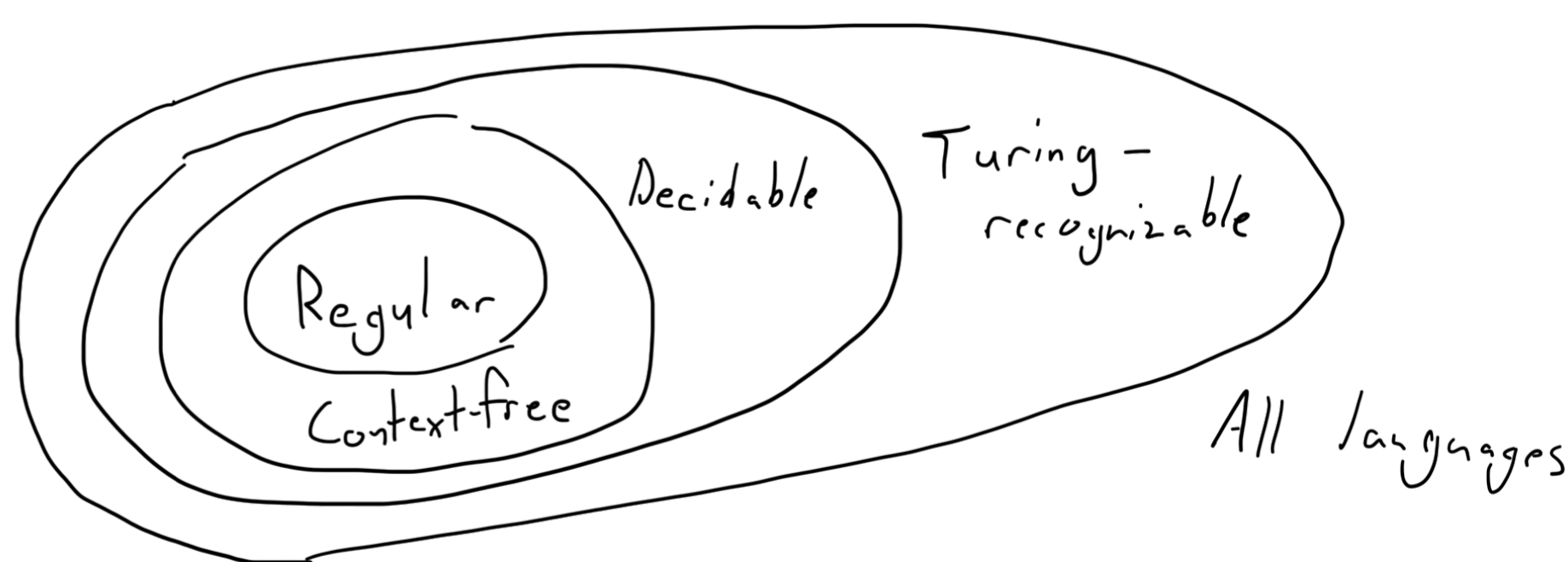


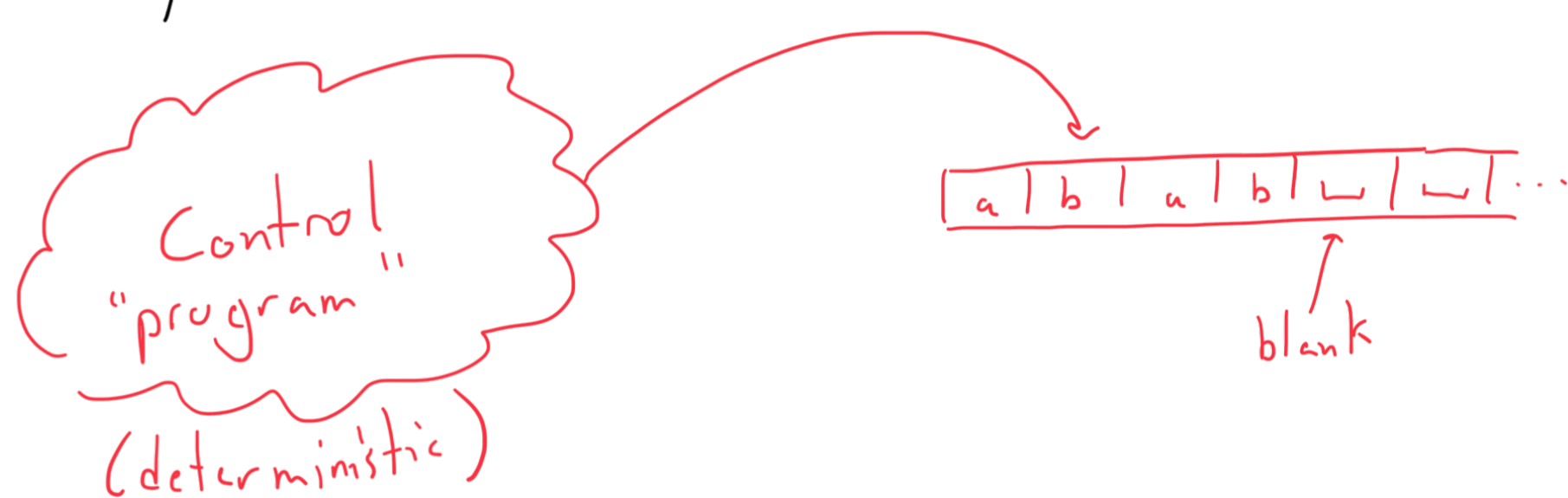
# Ch. 3 Church-Turing Thesis



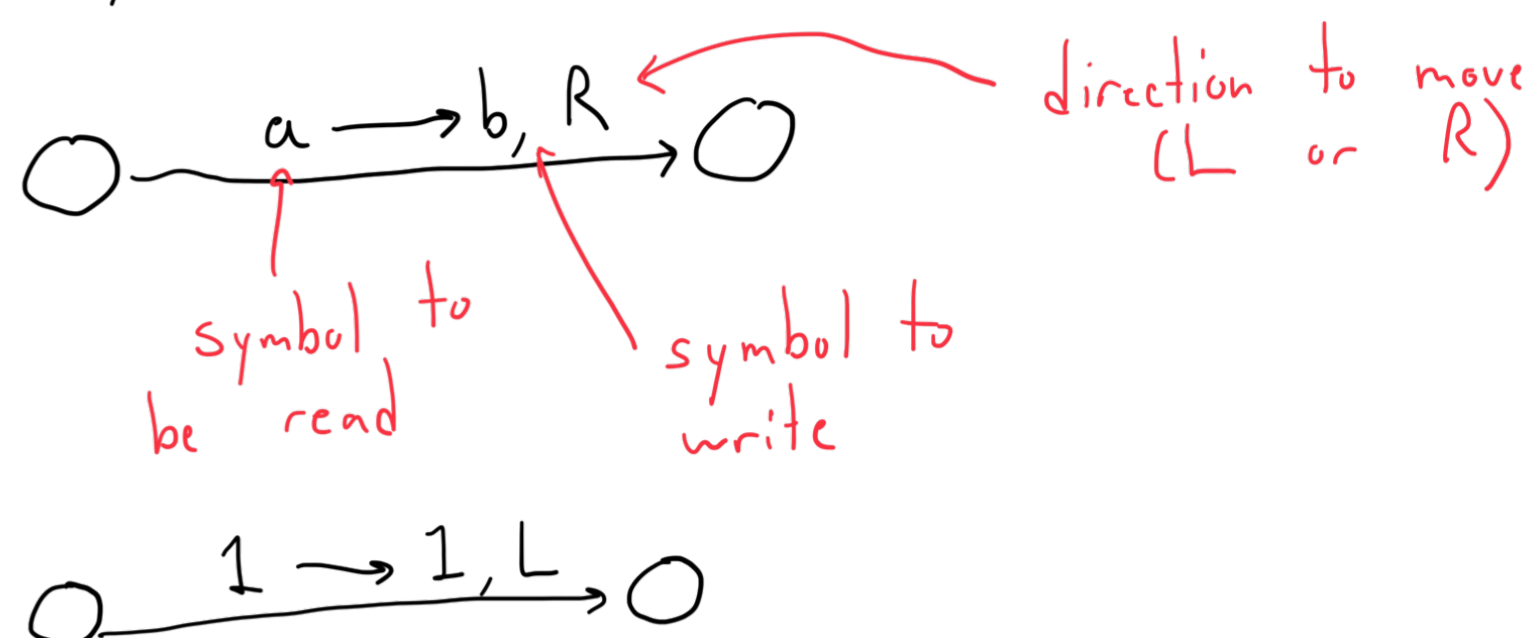
1930's: What does "computable" mean?  
 Alan Turing was the PhD student of  
 Alonzo Church. They concluded that

"Algorithmically computable" equals  
 "Computable by a Turing Machine (TM)"

A TM uses an infinite tape initially  
 containing only the input string and  
 blank everywhere else.



- Rules of operation) At each step of computation:
1. Read current symbol
  2. Update (i.e. write) to the same index
  3. Move exactly one index (cell) either left or right. If we are at the left-most index and try to move left, do not move.



Formal Definition A Turing Machine (TM) is a

7-tuple  $(Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$  where  
 $Q, \Sigma, \Gamma$  are all finite states and:

1.  $Q$  is a set of states
2.  $\Sigma$  is the input alphabet, not including the blank symbol  $\sqcup$
3.  $\Gamma$  is the tape alphabet, where  $\sqcup \in \Gamma$  and  $\Sigma \subseteq \Gamma$
4.  $\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R\}$  is the transition function.
5.  $q_0 \in Q$  is the start state.
6.  $q_{\text{accept}} \in Q$  is the accept state.
7.  $q_{\text{reject}} \in Q$  is the reject state where  $q_{\text{reject}} \neq q_{\text{accept}}$ .

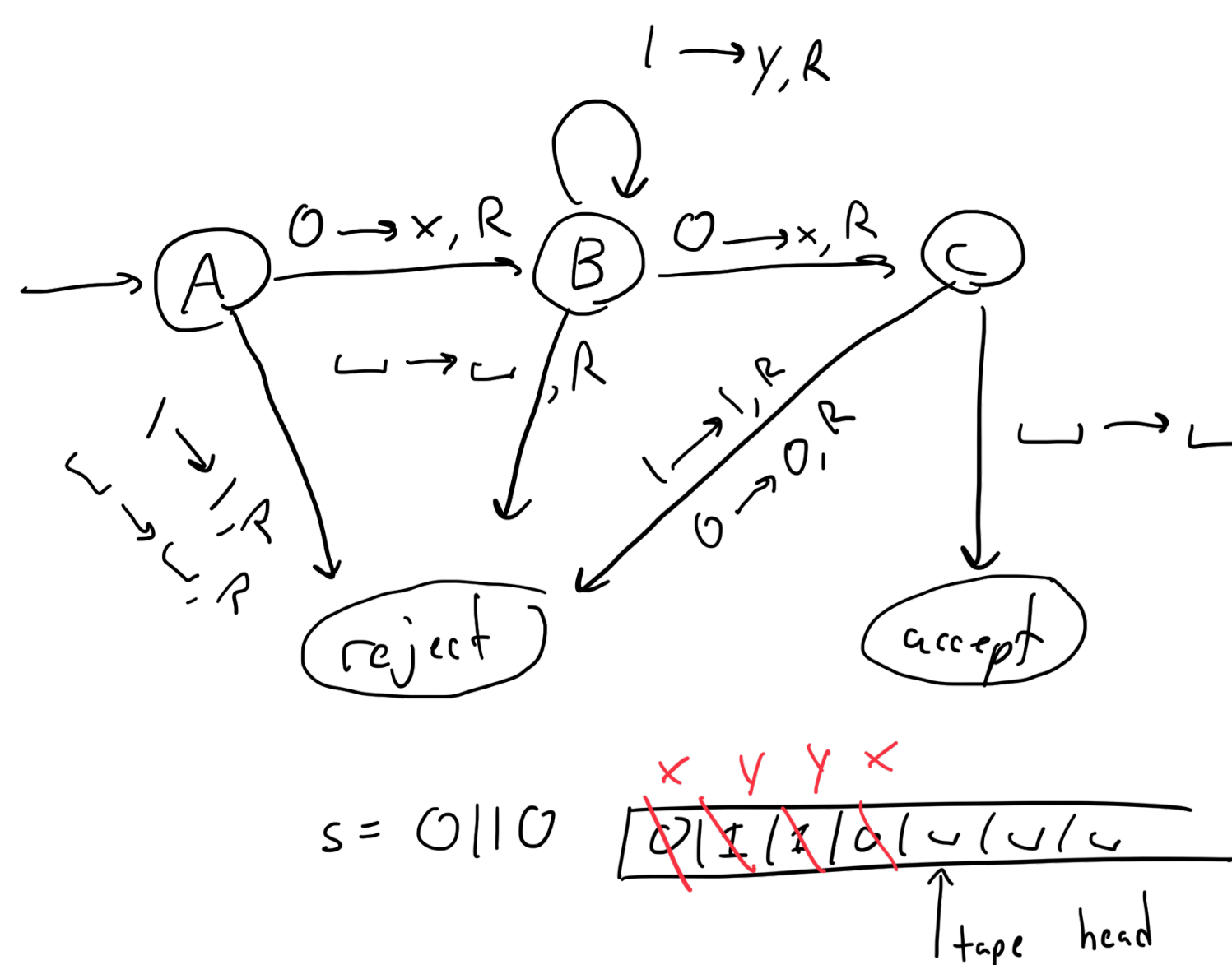
Computation can ...

- Halt and accept. When a TM enters the accept state, computation immediately halts.
- Halt and reject (... enters a reject state...)
- Loop. It fails to halt.

Given language  $L$ , if a TM "halts & accepts" on all  $s \in L$  and "halts & rejects" on all  $s \notin L$ , then  $L$  is decidable and the TM is a "decider."

Given language  $L$ , if a TM always halts and accepts when  $s \in L$ , then  $L$  is Turing-recognizable. And, the TM is a "recognizer."

Ex  $L = 01^*0$



Any string in  $L$  leads to a "halt & reject" or "halt & accepts", so the TM is a decider. This proves  $L$  is decidable (because at least one TM decides  $L$ ).

How do you prove a language is decidable?  
 Construct some TM that is a decider!

So, the TM "halts and accepts" on every  $s \in L$  and "halts and rejects" on every  $s \notin L$ .

How do you prove a language is Turing-recognizable?  
 Construct some TM that is a recognizer!

So, the TM "halts and accepts" on every  $s \in L$ .